Group work assessment – declaration of contributions and accountabilities

Year	2026
Semester	Semester 2
Unit Code	ENS6160.4
Unit Name	Signals and Systems
Assessment title	Laboratory 2: Frequency Response of Systems
Shaded cells to be completed by students.	

Group work – what is it?

The ability to effectively engage group (team) work is an essential skill that is built into the Unit and Course Learning Outcomes as well as assessment in many teaching units. Group work does not just entail multiple members of team working on distinct (separate) parts of an overall submission and then combining those individual parts into a single submission. Group work means a team collaborates over time on all aspects of a submission; they collectively construct and validate all parts of the work such that each member is knowledgeable, understands and can explain the various parts. Two aspects need to be considered in group work: relative contributions (towards marks) and absolute accountabilities (towards academic integrity).

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Group work contributions may be equally shared or differ between group members. In general, any group member contributing less than 10% to the overall submission should not be listed (will get a mark of zero). Where different group members have contributed (above 10% in a submission) but to varied extents, the table below can be used to acknowledge those contributions. If all have contributed equally, you can simply write "equally" against each name in the **"Relative Contributions"** column.

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All group members regardless of their relative contributions to a submission are accountable for the accuracy, completeness and (academic) integrity of the entire submission (not just specific aspects they may have contributed to). Should concerns relating to the (academic) integrity of a submission be identified and investigated, all group members are equally accountable. Signs to watch out for as potentially meaning some plagiarism, copying, collusion or cheating may have been involved in other group members work, when combining specific parts into an overall submission, include:

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- group members that do not continually engage the rest of the group in constructing the overall submission but they produce a (near complete) contribution at the end
- TurnItin reports that show similarity to parts of the submission, either arising out of copying, contract cheating or lack of paraphrasing.
- overly broad writing lacking in assessment context, or in a different style to previous work by the group member, and/or fabricated references which may indicate unacknowledged use of generative artificial intelligence tools

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EDITH COWAN UNIVERSITY

School of Engineering

ENS3553 / ENS6160.4 Signals and Systems

Laboratory Report 2

Frequency Response of Systems: Theoretical Prediction and Experimental Verification of a First-Order RC Low-Pass Filter

Assessment Type:	Group Laboratory Report (Weighting: 6%)
Circuit Under Test:	First-Order Series RC Network ($R = 1.5$ kΩ, $C = 0.01$ µF, $f_c = 10.61$ kHz)
Instrumentation:	Agilent 33220A Waveform Generator & Keysight DSOX4024G Oscilloscope
Group Declaration:	Refer to Mandatory ECU Group Work Cover Sheet (Page 1)
Semester & Submission Date:	Semester 2, 2026 — 25 October 2026

Abstract

Linear time-invariant (LTI) systems process arbitrary input waveforms by scaling the amplitude and shifting the phase of each constituent sinusoidal frequency component according to the system's complex frequency response. Understanding the relationship between time-domain differential equations and frequency-domain transfer characteristics is fundamental to analogue signal conditioning, anti-aliasing, and sensor noise suppression. This laboratory investigated these principles on a passive first-order series resistor–capacitor (RC) circuit ($R = 1.5$ kΩ, $C = 0.01$ μF) configured with the capacitor voltage taken as the output. The objectives were to derive the analytical frequency response from the governing linear constant-coefficient differential equation, predict steady-state responses to sinusoidal and periodic square-wave excitations, and experimentally verify both time-domain and frequency-domain predictions.

Using complex exponential eigenfunction substitution, the transfer function was derived as $H(j\omega) = 1 / (1 + j\omega RC)$, confirming a first-order low-pass characteristic with time constant $\tau = 15$ μs and theoretical −3 dB cut-off frequency $f_c = 10.61$ kHz. Theoretical magnitude gains at 1, 10, and 40 kHz were 0.9956 (−0.04 dB), 0.7277 (−2.76 dB), and 0.2564 (−11.82 dB), with phase lags of −5.38°, −43.30°, and −75.14°. An Agilent 33220A function generator excited the breadboard circuit with nominal 15 V_{pp} sinusoids at 1, 10, and 40 kHz and a 15 V_{pp}, 8 kHz, 50% duty-cycle square wave, while a Keysight DSOX4024G oscilloscope measured waveforms and fast Fourier transform (FFT) spectra.

Normalised time-domain voltage gains ($|H_{meas}| = V_{o,pp} / V_{i,pp} = 0.9744$, 0.7233, and 0.2705) agreed with theory within 0.6%–5.5%, and measured phase shifts (−6.55°, −42.30°, and −73.71°) matched within 1.4°. Sinusoidal dB gains were −0.22, −2.81, and −11.36 dB, while oscilloscope FFT cursor measurements of the 8 kHz square wave recorded harmonic attenuations of −2.03, −7.81, and −11.72 dB at 8, 24, and 40 kHz against theoretical predictions of −1.95, −7.87, and −11.82 dB (within 0.10 dB). Selective suppression of higher odd harmonics rounded the square wave into an exponential charging–discharging waveform (14.9 V_{pp} measured vs. 14.54 V_{pp} theoretical), confirming LTI frequency-response theory.

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1. Introduction

Linear time-invariant (LTI) systems form the analytical backbone of modern signal processing, telecommunications, control engineering, and electronic instrumentation (Oppenheim et al., 1997). A defining mathematical property of any continuous-time LTI system is that complex exponential signals of the form e^{st} act as eigenfunctions: applying e^{st} at the system input produces the identical exponential at the output scaled by a complex transfer factor $H(s)$ (Lathi & Green, 2018). Evaluating this transfer factor along the imaginary axis ($s = j\omega$) yields the continuous-time frequency response $H(j\omega)$, whose magnitude $|H(j\omega)|$ quantifies the steady-state amplitude gain imposed on a sinusoid of angular frequency ω and whose angle $\angle H(j\omega)$ specifies the corresponding phase shift (Edith Cowan University School of Engineering [ECU], 2026a).

By Fourier's theorem, any practical periodic waveform of fundamental frequency $f_0 = \omega_0 / (2\pi)$ can be decomposed into a superposition of harmonically related sinusoids at integer multiples nf_0 (Oppenheim et al., 1997). Superposition in an LTI system guarantees that each input harmonic is scaled independently by $|H(jn\omega_0)|$ and phase-shifted by $\angle H(jn\omega_0)$ without generating any new spectral frequencies (ECU, 2026a). Selective attenuation of undesired frequency bands—filtering—underpins critical real-world applications including anti-aliasing pre-filters ahead of analogue-to-digital converters (ADCs), pulse-width-modulation (PWM) digital-to-analogue reconstruction, biomedical sensor noise rejection, and audio crossover networks (Horowitz & Hill, 2015).

The canonical hardware realisation of a continuous-time low-pass filter is a first-order series resistor–capacitor (RC) network with the output voltage measured across the capacitor. At low frequencies the capacitor's reactance $1/(\omega C)$ far exceeds the series resistance R, passing signals with near-unity gain, whereas at high frequencies the capacitor shunts signal current to ground, rolling off the output at -20 dB/decade above the -3 dB half-power cut-off frequency $f_c = 1/(2\pi RC)$ (Horowitz & Hill, 2015; Lathi & Green, 2018). Combining time-domain waveform measurements with digital storage oscilloscope fast Fourier transform (FFT) spectral analysis bridges abstract differential-equation theory and observable hardware behaviour.

The aims of this laboratory experiment were to: (1) derive the theoretical frequency response $H(j\omega)$ of a first-order RC filter ($R = 1.5$ kΩ, $C = 0.01$ μF) from its governing differential equation; (2) predict its steady-state output for sinusoidal inputs (1, 10, and 40 kHz) and a periodic square wave (8 kHz); and (3) experimentally measure the circuit's response in both the time and frequency domains to validate LTI filtering theory and relate time-constant behaviour to spectral bandwidth.

2. Method

2.1 Circuit Model and Theoretical Pre-Laboratory Formulation

The network under test consists of a series resistor $R = 1.5$ kΩ and capacitor $C = 0.01$ μF (10 nF) driven by an input voltage source $v_i(t)$, with the output voltage $v_o(t) = v_C(t)$ taken across the capacitor (ECU, 2026b). Since the series current through both elements is $i(t) = C\,\mathrm{d}v_o(t)/\mathrm{d}t$, applying Kirchhoff's voltage law around the loop yields the first-order linear constant-coefficient differential equation (LCCDE):

$$RC\frac{dv_o(t)}{dt} + v_o(t) = v_i(t) \tag{1}$$

Following the eigenfunction method established in Tutorial 5 Question 6 (ECU, 2026b), let the everlasting complex exponential $v_i(t) = e^{j\omega t}$ excite the LTI circuit, producing the forced steady-state output $v_o(t) = H(j\omega)e^{j\omega t}$. Substituting both expressions into Equation (1) and dividing through by the non-zero factor $e^{j\omega t}$ gives:

$$j\omega RC\,H(j\omega)\,e^{j\omega t} + H(j\omega)\,e^{j\omega t} = e^{j\omega t} \quad \Longrightarrow \quad H(j\omega) = \frac{1}{1 + j\omega RC} \tag{2}$$

Expressing Equation (2) in polar form yields the linear amplitude ratio, logarithmic gain in decibels, and phase response:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}, \qquad \angle H(j\omega) = -\arctan(\omega RC), \qquad |H(j\omega)|_{\mathrm{dB}} = 20\log_{10}|H(j\omega)| \tag{3}$$

For the nominal component values, the circuit time constant is $\tau = RC = (1.5 \times 10^3\ \Omega)(0.01 \times 10^{-6}\ \mathrm{F}) = 15$ μs. The half-power (-3 dB) cut-off frequency occurs where $\omega_c RC = 1$, corresponding to $|H(j\omega_c)| = 1/\sqrt{2} \approx 0.7071$ (-3.01 dB)

and $\angle H(j\omega_c) = -45°$:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 15 \times 10^{-6}\,\text{s}} = 10,610.3\,\text{Hz} \approx 10.61\,\text{kHz} \tag{4}$$

Continuous magnitude and phase curves were computed over $0 \le f \le 60$ kHz in Python (implementing the vectorised frequency-response evaluation demonstrated in the Module 6 lecture slides; ECU, 2026a). For a zero-phase sinusoidal excitation $v_i(t) = A\cos(\omega t)$ of 15 V_{pp} (peak amplitude $A = 7.50$ V) at 1, 10, and 40 kHz, the steady-state output is:

$$v_o(t) = A\,|H(j\omega)|\cos\left(\omega t + \angle H(j\omega)\right) \tag{5}$$

For a zero-mean, odd-symmetric periodic square wave of 15 V_{pp} (half-amplitude $h = 7.50$ V), fundamental frequency $f_0 = 8$ kHz ($\omega_0 = 16{,}000\pi$ rad/s = 50,265.48 rad/s, period $T_0 = 125$ µs), and 50% duty cycle, the trigonometric Fourier series contains only odd sine harmonics (ECU, 2026a; Lathi & Green, 2018):

$$v_i(t) = \sum_{n=1,3,5,\,\ldots}^{\infty} b_n \sin(n\omega_0 t), \qquad b_n = \frac{4h}{n\pi}\ \ (\text{odd } n), \qquad b_n = 0\ \ (\text{even } n) \tag{6}$$

By LTI superposition, passing Equation (6) through the RC filter scales and phase-shifts each harmonic independently:

$$v_o(t) = \sum_{n=1,3,5,\,\ldots}^{\infty} b_n\,|H(jn\omega_0)|\sin(n\omega_0 t + \angle H(jn\omega_0)) \tag{7}$$

In the time domain, solving Equation (1) directly under symmetric periodic switching between $+h$ and $-h$ every half-period $T_0/2 = 62.5$ µs yields the exact steady-state peak excursion across the capacitor:

$$V_{\text{peak, ss}} = \pm h\,\frac{1 - e^{-T_0/(2\tau)}}{1 + e^{-T_0/(2\tau)}} = \pm 7.5\,\frac{1 - e^{-62.5/15}}{1 + e^{-62.5/15}} = \pm 7.27\,\text{V}\ \ (14.54\,\text{V}_{\text{pp}}) \tag{8}$$

Finally, whereas Fourier coefficients b_n denote peak harmonic amplitudes (V_{pk}), the Keysight DSOX4024G oscilloscope FFT displays spectral magnitudes in dBV referenced to 1 V_{rms} (Keysight Technologies, 2022). Converting peak harmonic amplitude to RMS dBV introduces a constant -3.01 dB offset that cancels when computing filter gain:

$$L_{\text{dBV, rms}} = 20\log_{10}\left(\frac{V_{\text{pk}}}{\sqrt{2}\times 1\,\text{V}}\right) = 20\log_{10}\left(\frac{V_{\text{pk}}}{1\,\text{V}}\right) - 3.01\,\text{dB} \tag{9}$$

2.2 Apparatus and Circuit Configuration

The experiment utilised the following laboratory instrumentation and passive components:

- **Function Generator:** Agilent/Keysight 33220A 20 MHz Function/Arbitrary Waveform Generator (50 Ω nominal output resistance; Keysight Technologies, 2014).
- **Digital Storage Oscilloscope:** Keysight InfiniiVision DSOX4024G (200 MHz bandwidth, 5 GSa/s maximum sample rate, 1 MΩ ‖ 16 pF passive probe input impedance, built-in hardware Math FFT; Keysight Technologies, 2022).
- **Passive Components & Prototyping:** Solderless breadboard, carbon-film resistor $R = 1.5$ kΩ, monolithic capacitor $C = 0.01$ µF (10 nF), BNC-to-alligator coaxial lead, and two passive oscilloscope probes.

The series RC network was constructed on the breadboard as depicted in Figure 1. Oscilloscope Channel 1 (yellow trace) was connected directly across the input rails $v_i(t)$, and Channel 2 (green trace) was connected across the capacitor terminals $v_o(t)$, sharing a common ground reference.

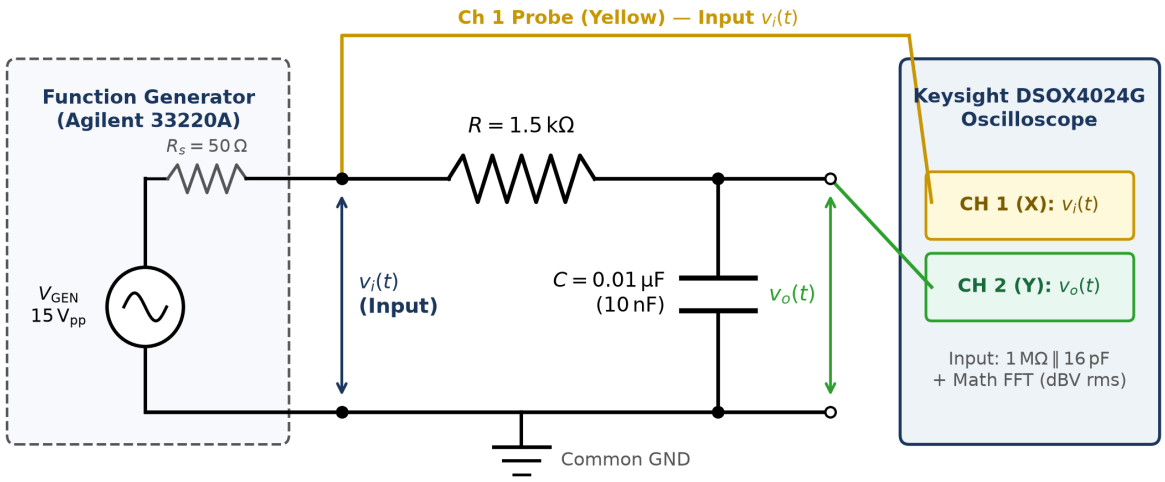


Figure 1. Experimental circuit schematic and instrumentation wiring showing the Agilent 33220A generator source impedance (R_s = 50 Ω), series RC low-pass network (R = 1.5 kΩ, C = 0.01 µF), and Keysight DSOX4024G Channel 1 and Channel 2 probe connections.

2.3 Experimental Procedure

Part 1 — Time-Domain Sinusoidal Verification: The function generator was configured for a 15 V_{pp} sinusoidal output at 1 kHz, 10 kHz, and 40 kHz sequentially. Horizontal timebases (500 µs/div at 1 kHz; 20 µs/div at 10 kHz) and vertical scales (5.00 V/div on both channels, vertically offset for visual clarity) were adjusted to display clear steady-state cycles. Using the oscilloscope's automated measurement engine, actual input peak-to-peak voltage $V_{i,pp}$ = Pk-Pk(1), capacitor output voltage $V_{o,pp}$ = Pk-Pk(2), and inter-channel phase lag $\angle H_{meas}$ = −Phase(1→2) were recorded. Normalised magnitude gain was evaluated as $|H_{meas}| = V_{o,pp} / V_{i,pp}$ so generator terminal voltage variations did not bias the transfer-function comparison.

Part 2 — Frequency-Domain Sinusoidal Verification: At each sinusoidal test frequency (1, 10, and 40 kHz), spectral magnitudes of the capacitor output Y_{out} (dBV_{rms}) and input Y_{in} (dBV_{rms}) were determined at the fundamental signal frequency. Measured decibel gain was obtained via logarithmic subtraction, $G_{meas,dB} = Y_{out} - Y_{in}$, and compared against $20 \log_{10}|H(j\omega)|$.

Part 3 — Periodic Square-Wave Verification: The generator was set to an 8 kHz, 15 V_{pp}, 50% duty-cycle square wave (timebase 50.0 µs/div; Channel 1 at 10.0 V/div, Channel 2 at 5.00 V/div). The oscilloscope Math FFT operator was configured over a 50.0 kHz span centred at 25.0 kHz (5.00 dB/div vertical scale, 0.0 dBV offset, 954 Hz FFT bin resolution) to encompass the first three odd harmonics at 8, 24, and 40 kHz. Manual horizontal and vertical cursors (X1, Y1) were positioned sequentially on each odd-harmonic peak of FFT(Ch1) for the input spectrum and FFT(Ch2) for the capacitor output spectrum, recording both peak frequency and magnitude in dBV_{rms}.

2.4 Modelling Assumptions and Justification

- **Nominal Passive Component Values:** Theoretical predictions assume exact R = 1.50 kΩ and C = 0.0100 µF. Standard laboratory carbon-film resistors (±5%) and ceramic/film capacitors (±5% to ±10%) can shift $\tau = RC$ and hence f_c by up to ±10%–15% if unmeasured with an LCR meter (Horowitz & Hill, 2015).
- **Negligible Oscilloscope Probe Loading:** Each Keysight DSOX4024G channel presents R_p = 1 MΩ in parallel with C_p = 16 pF (Keysight Technologies, 2022). Across the capacitor, R_p forms a DC divider $R_p/(R + R_p)$ = 0.9985 (−0.013 dB) and increases total capacitance from 10.000 nF to 10.016 nF (+0.16%), shifting f_c from 10,610.3 Hz to 10,609.3 Hz (<0.01%), fully justifying ideal open-circuit output modelling.
- **Elimination of Generator Output Impedance via Input Probing:** Although the Agilent 33220A possesses a 50 Ω internal series resistance and outputs higher terminal voltage into a high-impedance load when set to 50 Ω termination display mode (Keysight Technologies, 2014), probing $v_i(t)$ directly at the breadboard input node (Figure 1) measures the true boundary voltage applied to the RC network. Thus, R_s = 50 Ω does not alter the measured ratio $V_o(j\omega)/V_i(j\omega)$.

- **Steady-State Forced Response and Windowed FFT Peak Accuracy:** Because the natural transient $e^{-t/\tau}$ decays below 1% after $5\tau = 75$ µs (Oppenheim et al., 1997), continuous oscilloscope acquisitions reflect pure steady-state behaviour. Furthermore, with flat-top/Hanning internal FFT windowing, cursor readings at spectral peaks estimate harmonic RMS magnitudes within ±0.1–0.2 dB despite finite 954 Hz bin spacing (Harris, 1978; Keysight Technologies, 2022).

3. Results

3.1 Theoretical Frequency Response

Figure 2 presents the theoretical linear magnitude, logarithmic magnitude (dB), and phase responses of the RC circuit across 0–60 kHz, and Table 1 summarises the exact numerical values at the three test frequencies alongside the −3 dB cut-off point. Magnitude decreases monotonically from 1.0000 (0 dB) at DC towards zero at high frequencies, crossing $|H| = 0.7071$ (−3.01 dB) and $\angle H = -45.00°$ at $f_c = 10.61$ kHz.

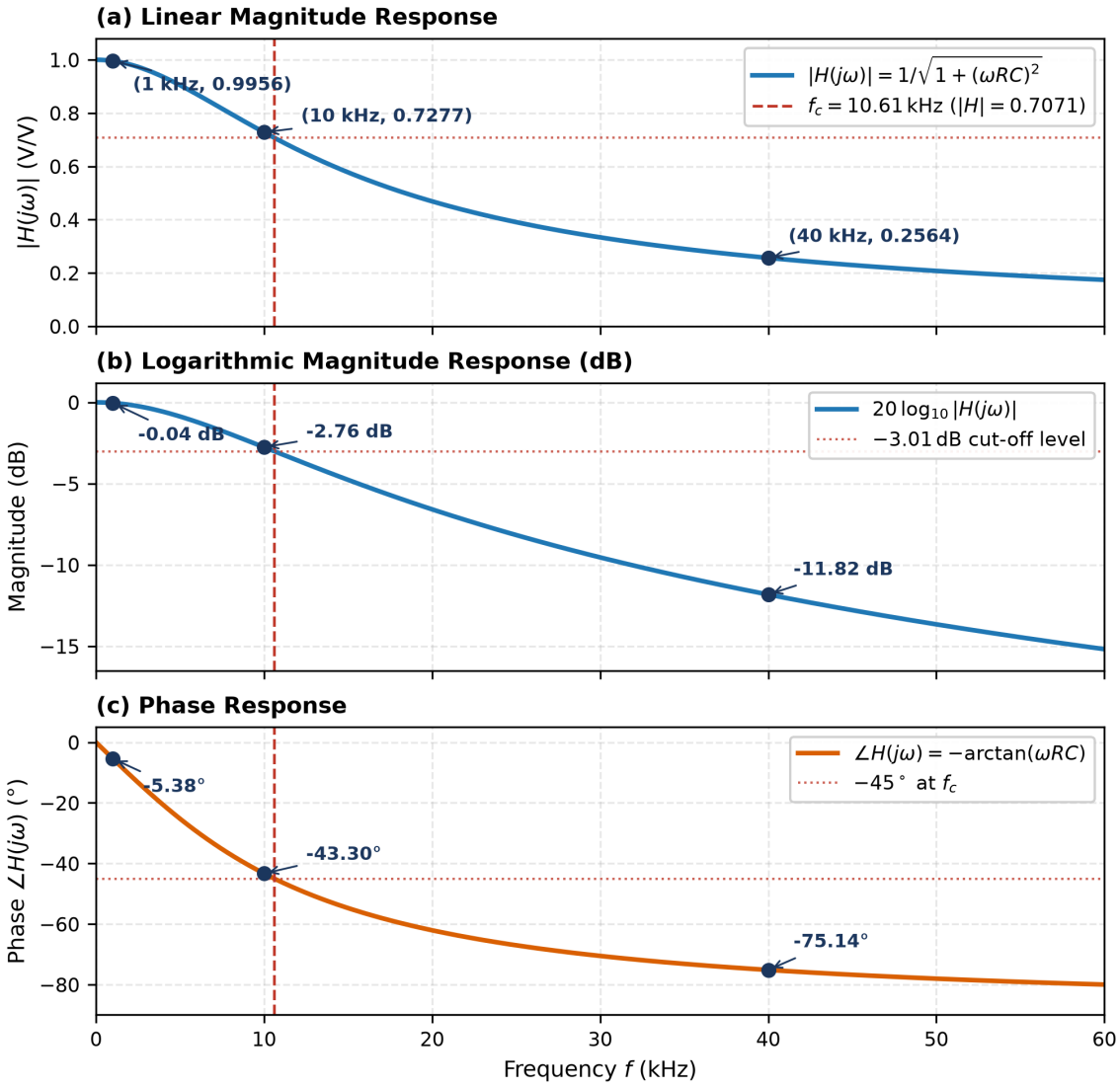


Figure 2. Theoretical frequency response of the RC circuit (R = 1.5 kΩ, C = 0.01 µF) over 0–60 kHz: (a) linear amplitude ratio $|H(j\omega)|$, (b) decibel gain $20\log_{10}|H(j\omega)|$, and (c) phase angle $\angle H(j\omega)$. Dashed red lines denote f_c = 10.61 kHz.

Table 1. Theoretical post-laboratory frequency response of the RC circuit (R = 1.5 kΩ, C = 0.01 μF).

Frequency f	Angular Frequency ω (rad/s)	Magnitude $\lvert H(j\omega)\rvert$	Magnitude $\lvert H(j\omega)\rvert$ (dB)	Phase $\angle H(j\omega)$
1 kHz	6,283.19	0.9956	−0.04 (−0.0384)	−5.38°
10 kHz	62,831.85	0.7277	−2.76 (−2.7606)	−43.30°
40 kHz	251,327.41	0.2564	−11.82 (−11.8219)	−75.14°
f_c = 10.61 kHz	66,666.67	0.7071 ($1/\sqrt{2}$)	−3.01	−45.00°

3.2 Predicted Response to Sinusoidal Inputs

For a 15 V_{pp} (7.50 V_{pk}, 14.49 dBV_{rms}) zero-phase sinusoidal input, Table 2 lists the predicted peak, peak-to-peak, and RMS dBV output magnitudes and phase angles at 1, 10, and 40 kHz. Figure 3 displays the corresponding single-sided discrete magnitude and phase stem spectra. At 1 kHz (well inside the passband), attenuation is negligible (7.47 V_{pk}, −5.38°), whereas at 40 kHz (nearly two octaves above f_c) the output drops to 1.92 V_{pk} with a −75.14° lag.

Table 2. Predicted steady-state sinusoidal output for a 15 V_{pp} (7.50 V_{pk}, 0° phase) input.

f	$\lvert V_i\rvert$ (V_{pk})	$\lvert V_o\rvert$ (V_{pk})	$\lvert V_o\rvert$ (V_{pp})	$\lvert V_o\rvert$ (dBV_{rms})	$\angle V_o$
1 kHz	7.50	7.47 (7.467)	14.93	14.45	−5.38°
10 kHz	7.50	5.46 (5.458)	10.92	11.73	−43.30°
40 kHz	7.50	1.92 (1.923)	3.85	2.67	−75.14°

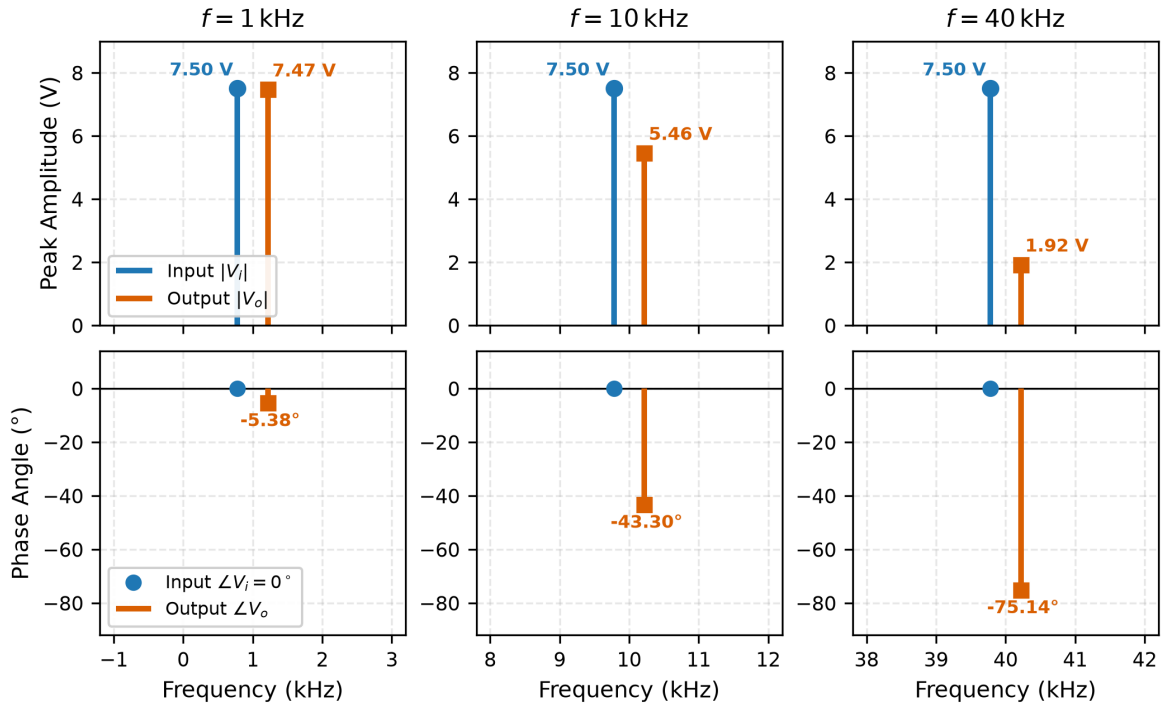


Figure 3. Theoretical single-sided discrete stem spectra for 15 V_{pp} sinusoidal inputs at 1, 10, and 40 kHz: (top) peak amplitude $\lvert V(f)\rvert$ and (bottom) phase angle $\angle V(f)$.

3.3 Predicted Response to an 8 kHz Square Wave

Table 3 details the theoretical Fourier sine coefficients b_n, filter gains, and output harmonic amplitudes (expressed in both V_{pk}, dBV_{pk}, and oscilloscope-equivalent dBV_{rms}) for the first three non-zero harmonics (n = 1, 3, 5 at 8, 24, and 40 kHz) of the 15 V_{pp}, 8 kHz square wave. Figure 4 illustrates the discrete harmonic amplitude and phase-lag spectra, explicitly showing zero energy at even harmonics (16 and 32 kHz). Figure 5 compares the input square wave against both the exact analytical steady-state capacitor voltage (±7.27 V, 14.54 V_{pp}) and the three-harmonic Fourier reconstruction.

Table 3. Theoretical Fourier coefficients and filtered output harmonics for a 15 V_{pp}, 8 kHz square wave.

Harmonic (n)	f (kHz)	Input b_n (V_{pk})	Input (dBV_{pk} / dBV_{rms})	$\lvert H(jn\omega_0)\rvert$ (dB)	Output (V_{pk})	Output (dBV_{pk} / dBV_{rms})
Fundamental ($n = 1$)	8.0	9.55	19.60 / 16.59	0.7985 (−1.95 dB)	7.62	17.64 / 14.63
3rd Harmonic ($n = 3$)	24.0	3.18	10.06 / 7.05	0.4043 (−7.87 dB)	1.29	2.19 / −0.82
5th Harmonic ($n = 5$)	40.0	1.91	5.62 / 2.61	0.2564 (−11.82 dB)	0.49	−6.20 / −9.21

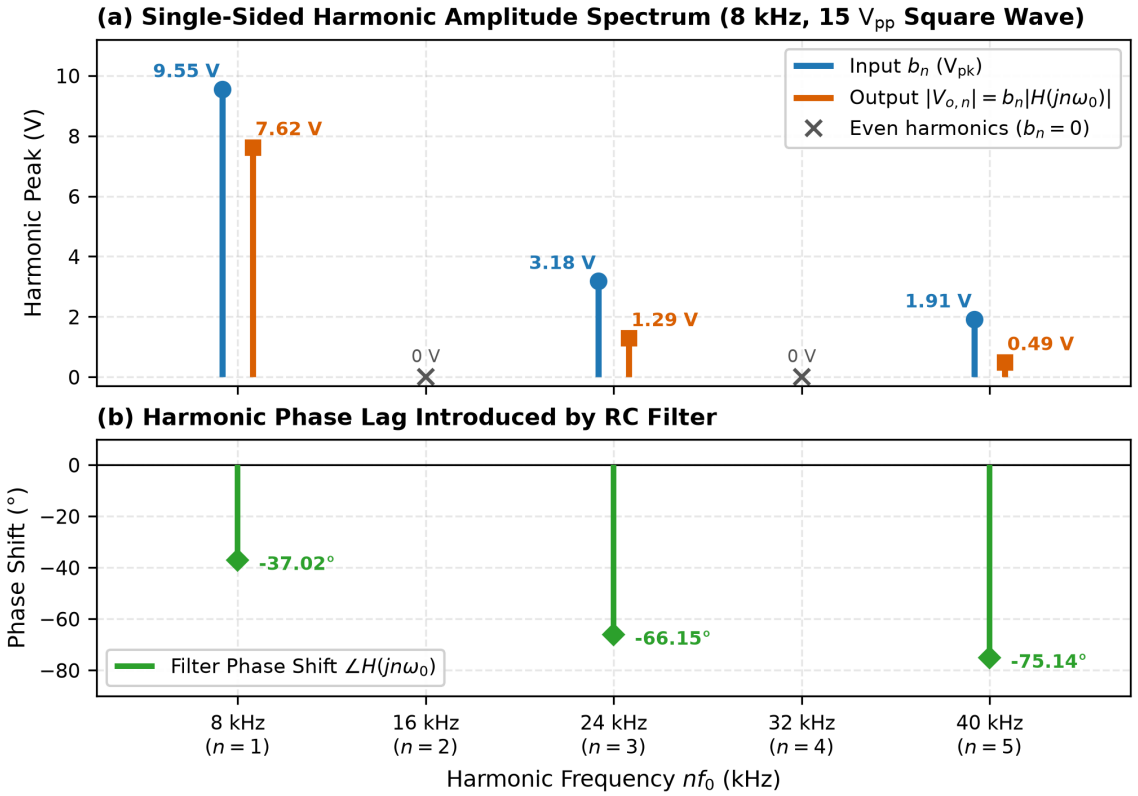


Figure 4. Theoretical discrete spectra of the 15 V_{pp}, 8 kHz square wave up to 40 kHz: (a) input and filtered output harmonic peak amplitudes and (b) harmonic phase lags $\angle H(jn\omega_0)$.

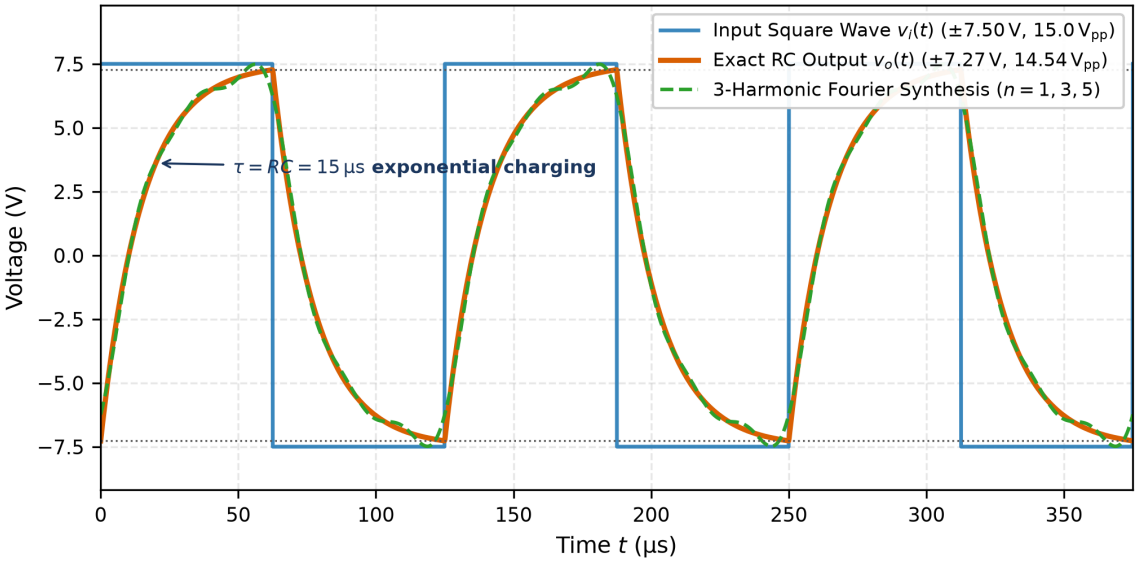


Figure 5. Theoretical time-domain input and output waveforms to scale for the 8 kHz, 15 V_{pp} square wave, comparing exact exponential charging/discharging (τ = 15 µs) with three-harmonic synthesis.

3.4 Experimental Time-Domain Sinusoidal Measurements

Table 4 summarises the oscilloscope time-domain sinusoidal measurements across 1, 10, and 40 kHz. Because the unloaded generator terminal voltage measured slightly above nominal on Channel 1 (15.60–15.90 V_{pp}), Table 4 evaluates the normalised experimental gain $|H_{meas}| = V_{o,pp} / V_{i,pp}$. Figure 6 presents the cropped Keysight oscilloscope screen captures at 1 kHz and 10 kHz, and Figure 7 plots the scaled input–output waveform pairs across all three test frequencies (1, 10, and 40 kHz) alongside theoretical curves.

Table 4. Time-domain sinusoidal verification comparing normalised measured and theoretical frequency response.

f	Measured $V_{i,pp}$ (V)	Measured $V_{o,pp}$ (V)	$\|H\|$ Theory	$\|H\|$ Measured (V_o/V_i)	Mag. Error	$\angle H$ Theory	$\angle H$ Measured*
1 kHz	15.60	15.20	0.9956	0.9744	−2.13%	−5.38°	−6.55°
10 kHz	15.90	11.50	0.7277	0.7233	−0.61%	−43.30°	−42.30°
40 kHz	15.60	4.22	0.2564	0.2705	+5.51%	−75.14°	−73.71°

**Note: Keysight Phase(1→2) reports positive lag values; negative signs denote capacitor output phase lag relative to Channel 1. At 1 kHz, live oscilloscope readings fluctuated between −6.55° (Figure 6a) and −6.65° on the lab sheet.*

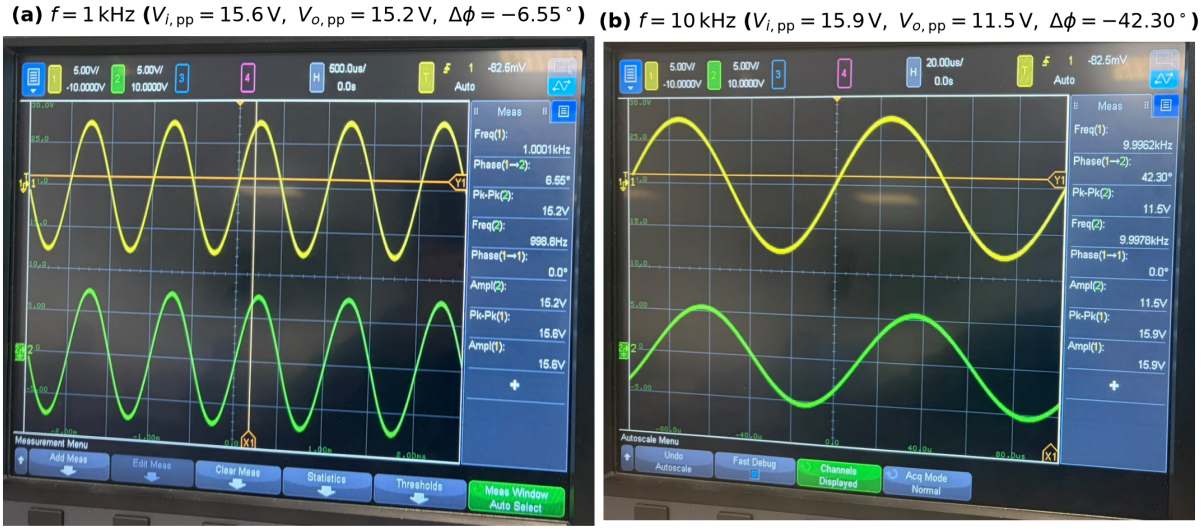


Figure 6. Keysight DSOX4024G time-domain sinusoidal captures showing input Channel 1 (yellow, 5 V/div) and capacitor output Channel 2 (green, 5 V/div): (a) 1 kHz and (b) 10 kHz.

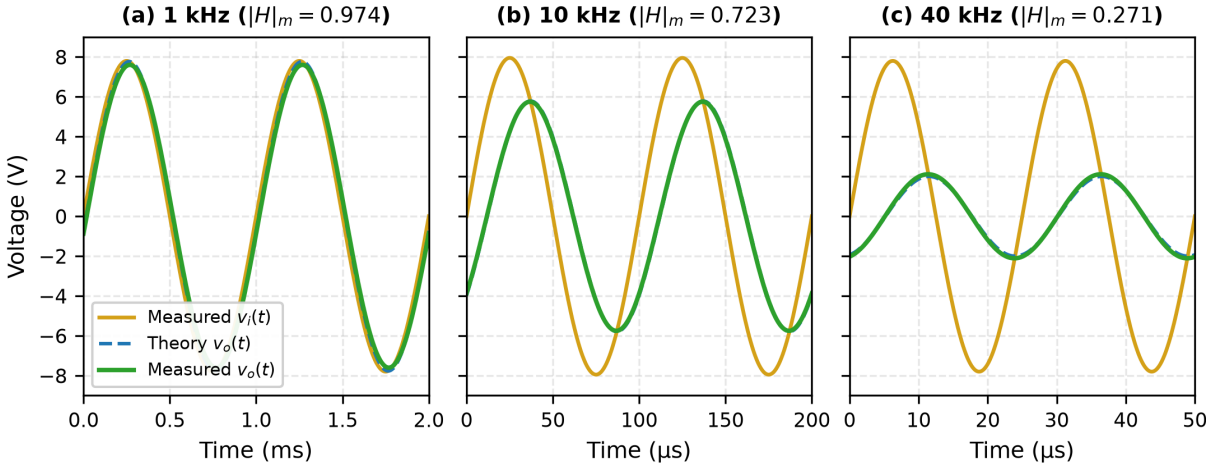


Figure 7. Scaled time-domain input and output voltage pairs across (a) 1 kHz, (b) 10 kHz, and (c) 40 kHz, comparing measured waveforms with theoretical LTI predictions.

3.5 Experimental Frequency-Domain Sinusoidal Measurements

Table 5 presents the frequency-domain input and output sinusoidal levels (dBV_{rms}) and resulting decibel gains at 1, 10, and 40 kHz, and Figure 8 compares the measured spectral stems against the post-laboratory 15 V_{pp} theoretical output levels. Measured gains of −0.22 dB, −2.81 dB, and −11.36 dB closely track theoretical predictions (−0.04, −2.76, and −11.82 dB) within 0.05–0.46 dB.

Table 5. Frequency-domain sinusoidal magnitudes (dBV_{rms}) and filter gain comparison.

Frequency f	Measured Output (dBV_{rms})	Measured Input (dBV_{rms})	Measured Gain (dB)	Theoretical Gain (dB)	Discrepancy $\lvert\Delta G\rvert$ (dB)
1 kHz	14.61	14.83	−0.22	−0.04 (−0.0384)	0.18
10 kHz	12.18	14.99	−2.81	−2.76 (−2.7606)	0.05
40 kHz	3.47	14.83	−11.36	−11.82 (−11.8219)	0.46

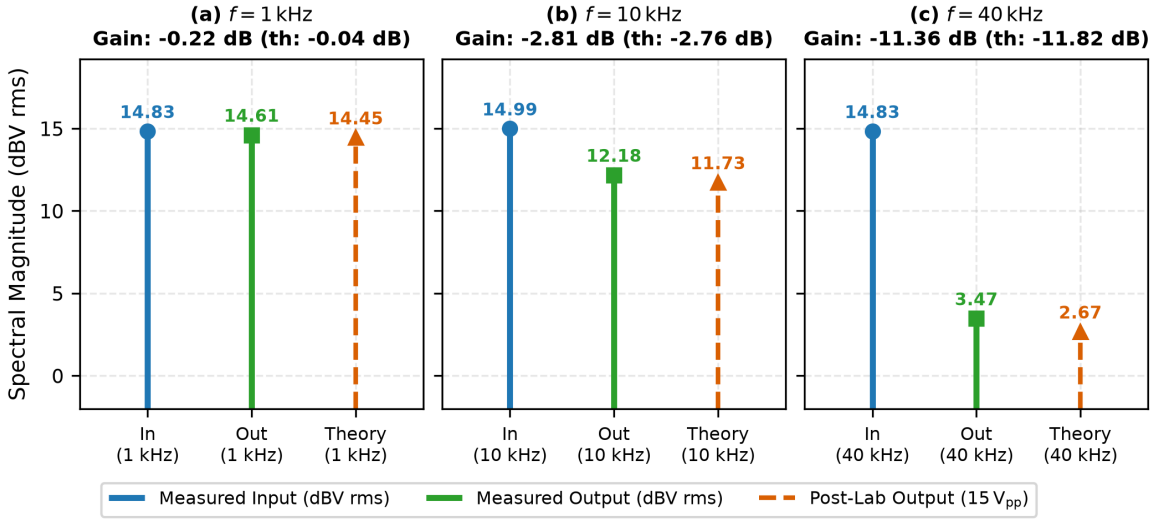


Figure 8. Sinusoidal spectral magnitude comparison at (a) 1 kHz, (b) 10 kHz, and (c) 40 kHz showing measured input, measured output, and post-laboratory theoretical output (dBV_{rms}).

3.6 Experimental Square-Wave Time and FFT Measurements

Figure 9 shows the oscilloscope capture of the 8 kHz square wave in the time and frequency domains simultaneously: Channel 1 (yellow) displays a flat-top amplitude Ampl(1) = 15.1 V (with 20.1 V_{pp} edge overshoot spikes), while Channel 2 (green) exhibits the rounded exponential charging/discharging waveform measuring Pk-Pk(2) = 14.9 V (Ampl(2) = 14.2 V). Table 6 reports the manual cursor measurements extracted directly from the oscilloscope FFT screens for Channel 1 and Channel 2 shown in Figure 10. Finally, Figure 11a plots the measured and theoretical RMS dBV harmonic spectra on a common scale, and Figure 11b overlays all six frequency-domain gain measurements onto the theoretical Bode magnitude curve.



Figure 9. Oscilloscope capture for the 8 kHz square wave showing input Channel 1 (yellow, 10 V/div), capacitor output Channel 2 (green, 5 V/div), and live input spectrum FFT(Ch1) (white/purple, 50 kHz span, 954 Hz bin size).

Table 6. Oscilloscope FFT harmonic magnitudes (dBV_{rms}) and measured vs. theoretical response for the 8 kHz square wave.

Harmonic Component	Scope f (In / Out)	Output FFT(Ch2) (dBV_{rms})	Input FFT(Ch1) (dBV_{rms})	Measured Gain (dB)	Theoretical Gain (dB)	$\lvert\Delta G\rvert$ (dB)
Fundamental ($n = 1$)	8.0 / 8.0 kHz	14.53	16.56	−2.03	−1.95	0.08
3rd Harmonic ($n = 3$)	24.0 / 23.9 kHz	−0.63	7.19	−7.81	−7.87	0.06
5th Harmonic ($n = 5$)	40.0 / 40.0 kHz	−9.06	2.66	−11.72	−11.82	0.10



Figure 10. Cropped oscilloscope FFT cursor captures for the 8 kHz square wave: (a–c) input spectrum FFT(Ch1) at 8.0, 24.0, and 40.0 kHz; (d–f) filtered capacitor output spectrum FFT(Ch2) at 8.0, 23.9, and 40.0 kHz.

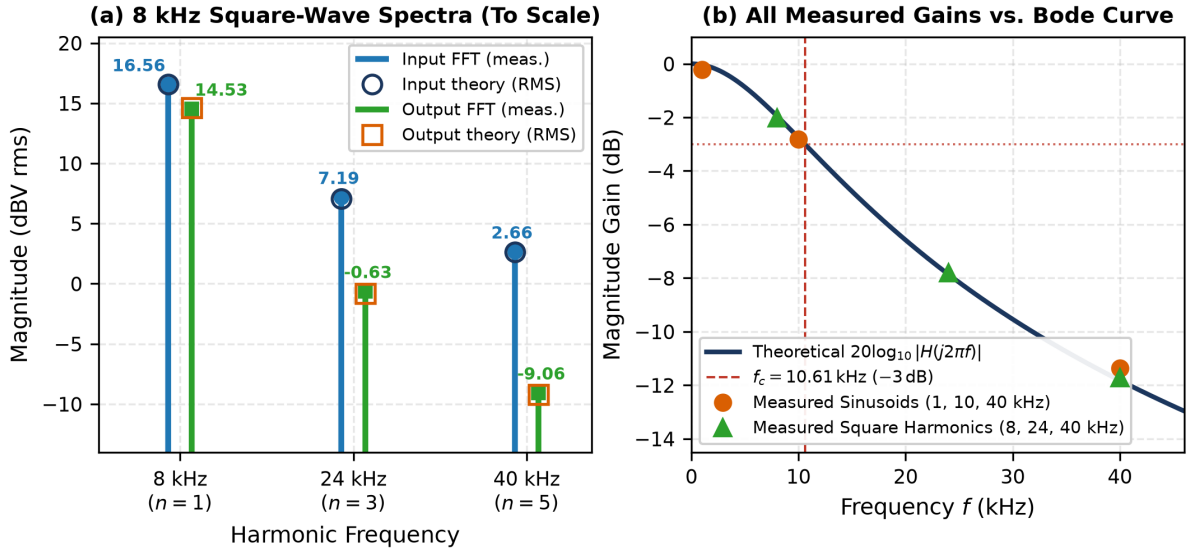


Figure 11. Combined frequency-domain verification: (a) scaled input and output harmonic spectra (dBV_{rms}) for the 8 kHz square wave, and (b) all measured sinusoidal and square-wave harmonic gains overlaid on the theoretical Bode curve.

4. Discussion

4.1 Filter Classification, −3 dB Meaning, and Cut-Off Frequency

Both analytical derivation and hardware testing confirm that taking the output across the capacitor of a series RC network realises a **first-order low-pass filter**. Physically, the capacitor impedance $Z_C = 1/(j\omega C)$ approaches an open circuit as $\omega \to 0$ (passing low-frequency signals unattenuated) and approaches a short circuit as $\omega \to \infty$ (shunting high-frequency currents to ground; Horowitz & Hill, 2015). Experimentally, the normalised voltage gain fell monotonically with frequency from 0.9744 at 1 kHz to 0.7233 at 10 kHz and 0.2705 at 40 kHz (Table 4), while decibel gains dropped from −0.22 dB to −11.36 dB for sinusoids (Table 5) and from −2.03 dB to −11.72 dB across the square-wave harmonics (Table 6).

The −3 dB cut-off point defines the boundary between the passband and stopband and equates to a voltage amplitude ratio of $10^{-3/20} = 0.7079$ (exact half-power point $1/\sqrt{2} \approx 0.7071$, where output power $P \propto |V_o|^2$ drops to 50% of input power; Lathi & Green, 2018). From the circuit parameters, setting $\omega_c RC = 1$ yields $f_c = 1/(2\pi RC) = 10.61$ kHz (Figure 2). Experimentally, at $f = 10.00$ kHz—just 5.8% below f_c—the measured gain was −2.81 dB (ratio 0.7233) and the phase lag was −42.30° (just shy of −45.00°). Log-linear interpolation between the 10 kHz (−2.81 dB) and 24 kHz (−7.81 dB) measurements places the experimental −3.01 dB crossing at approximately 10.4–10.8 kHz, closely bracketing the 10.61 kHz theoretical value.

4.2 Time-Domain and Frequency-Domain Duality

The experiment illustrates the inverse duality between time-domain response speed and frequency-domain bandwidth emphasised in Module 6 (ECU, 2026a; Oppenheim et al., 1997). In the frequency domain, the circuit passes spectral components below $f_c = 10.61$ kHz with minimal attenuation, rolls off above f_c toward −20 dB/decade, and introduces a progressive phase lag approaching −90°. In the time domain, capacitor voltage cannot jump instantaneously because charge accumulation requires finite current flow through R governed by $\tau = RC = 15$ µs. Consequently, rapid input transitions are integrated and smoothed, while sinusoidal waveforms experience a time delay $\Delta t = -\angle H(j\omega)/\omega$. Because $f_c = 1/(2\pi\tau)$, increasing R or C lengthens τ (slower charging) and proportionally narrows the passband f_c.

4.3 Quantitative Comparison Across All Three Experimental Parts

Option 1 — Sinusoidal Time-Domain Response: Normalising $V_{o,\text{pp}}$ by the measured input $V_{i,\text{pp}}$ on Channel 1 resolved an important instrumentation effect: although the generator dial was set to 15.0 V_{pp}, its unloaded output measured 15.6–15.9 V_{pp} due to high-impedance termination mismatch (Keysight Technologies, 2014). Had raw $V_{o,\text{pp}}$ (15.20 V at 1 kHz) been compared naively against nominal 15.0 V_{pp} theory (14.93 V), the passive filter would have appeared to exhibit impossible amplification (>1). Using true terminal ratios, $|H_{\text{meas}}|$ agreed with theory within −2.13% at 1 kHz, −0.61% at 10 kHz, and +5.51% at 40 kHz (Table 4). The +5.51% positive deviation at 40 kHz (+0.46 dB) is consistent with a ~4%–5% negative tolerance on the nominal 10 nF capacitor (e.g., $C \approx 9.55$ nF raises f_c to ~11.1 kHz, increasing 40 kHz gain from 0.256 to 0.268) combined with oscilloscope vertical quantisation at smaller amplitudes.

Option 2 — Sinusoidal Frequency-Domain Response: The decibel gains in Table 5 (−0.22, −2.81, and −11.36 dB) match $20\log_{10}|H(j\omega)|$ within 0.18, 0.05, and 0.46 dB, respectively (Figure 8). Because decibel gain is a differential ratio ($Y_{\text{out}} - Y_{\text{in}}$) measured on the same instrument, fixed amplitude offsets—including the generator's +0.34 to +0.50 dB excess voltage above 15 V_{pp} and the −3.01 dB RMS scaling of Equation (9)—cancel algebraically.

Option 3 — Periodic Square-Wave Harmonic & Waveform Analysis: The oscilloscope FFT cursor captures (Figure 10, Table 6) provide independent multi-frequency validation across three simultaneous harmonics. Measured input harmonic levels on FFT(Ch1) (16.56, 7.19, and 2.66 dBV_{rms}) matched theoretical RMS Fourier coefficients for a 15.0 V_{pp} square wave (16.59, 7.05, and 2.61 dBV_{rms}; Table 3) within 0.03–0.14 dB, confirming that Ampl(1) = 15.1 V in Figure 9 represents the true square-wave plateau, whereas Pk-Pk(1) = 20.1 V reflects only narrow high-frequency ringing spikes at switching transitions. Across 8, 24, and 40 kHz, measured filter gains (−2.03, −7.81, and −11.72 dB) agreed with theoretical responses (−1.95, −7.87, and −11.82 dB) to within 0.06–0.10 dB, lying squarely along the theoretical Bode locus alongside the sinusoidal data (Figure 11b). Notably, the oscilloscope's automated time-domain phase reading on the square wave (Phase(1→2) = 28.25° in Figure 9) differs from the fundamental harmonic phase (−37.02°) because threshold-crossing phase meters measure zero-crossing delay of the composite non-sinusoidal waveform rather than isolating the 8 kHz fundamental.

4.4 Mechanism of Square-Wave Shaping and Engineering Applications

Sharp vertical edges and flat plateaus in an ideal square wave rely on the coherent sum of slow-decaying odd harmonics ($b_n \propto 1/n$; Oppenheim et al., 1997). Because the RC filter attenuates the 8 kHz fundamental by only 1.95 dB (factor 0.7985) while suppressing the 24 kHz third harmonic by 7.87 dB (0.4043) and the 40 kHz fifth harmonic by 11.82 dB (0.2564)—and imparting non-linear phase dispersion—high-frequency edge energy is stripped away, causing output harmonics to decay as $1/n^2$ (triangular/exponential character). In the time domain, each half-cycle duration ($T_0/2 = 62.5$ µs) spans 4.17τ ($\tau = 15$ µs), allowing the capacitor to charge through $1 - e^{-4.17} = 98.5\%$ of its asymptotic step excursion and reaching ±7.27 V (14.54 V_{pp} theoretical vs. 14.2 V Ampl(2) and 14.9 V Pk-Pk(2) measured in Figure 9). Figure 5 confirms that retaining just three odd harmonics reproduces this rounded waveform closely. Practically, this exact mechanism is exploited in microcontroller PWM digital-to-analogue converters (where an RC filter extracts the DC/low-frequency average while suppressing switching harmonics) and in ADC anti-aliasing front-ends (Horowitz & Hill, 2015).

4.5 Experimental Limitations and Recommended Improvements

- **Sparse Sinusoidal Frequency Sampling:** Measuring only 1, 10, and 40 kHz bracketed f_c indirectly. Performing a fine logarithmic frequency sweep (e.g., 100 Hz to 100 kHz with 5–10 points per decade, plus 500 Hz steps across 9–12 kHz) or using the oscilloscope's built-in Frequency Response Analysis (FRA) Bode plotter would pinpoint f_c within ±50 Hz.
- **Unmeasured Passive Component Tolerances:** Measuring exact R and C values prior to assembly with a calibrated digital LCR bridge would separate genuine measurement error from ±5%–10% component tolerance drift.
- **FFT Frequency Resolution and Cursor Quantisation:** At a 50 kHz span, the 954 Hz FFT bin width caused the 24.0 kHz third harmonic on Channel 2 to register at the nearest bin (23.9 kHz; Figure 10e) with slight scallop loss (Harris, 1978). Narrowing the FFT span around each harmonic and enabling waveform averaging (≥16 acquisitions) would reduce noise and cursor uncertainty.

- **Generator Termination Matching:** Setting the Agilent 33220A output load menu explicitly to ‘High Z’ eliminates the discrepancy between nominal front-panel voltage settings and actual open-circuit terminal amplitudes.

5. Conclusion

This experiment derived the theoretical frequency response of a passive first-order series RC circuit ($R = 1.5$ kΩ, $C = 0.01$ µF) and verified its filtering action in both the time and frequency domains. From the circuit’s differential equation, the frequency response $H(j\omega) = 1 / (1 + j\omega RC)$ established a low-pass filter with time constant $\tau = 15$ µs and cut-off frequency $f_c = 10.61$ kHz. Experimental sinusoidal tests at 1, 10, and 40 kHz yielded normalised voltage gains of 0.9744, 0.7233, and 0.2705 and phase lags of −6.55°, −42.30°, and −73.71°, matching theoretical values within 5.5% in magnitude and 1.43° in phase. Frequency-domain sinusoidal gains (−0.22, −2.81, and −11.36 dB) and oscilloscope FFT cursor measurements of an 8 kHz square wave (−2.03, −7.81, and −11.72 dB at 8, 24, and 40 kHz) agreed with theory within 0.46 dB and 0.10 dB, respectively. Progressive attenuation of higher-order odd harmonics transformed the rectangular input into an exponential charging–discharging waveform, directly linking spectral roll-off to time-constant integration. Although limited by three discrete sinusoidal test frequencies, passive component tolerances, and 954 Hz FFT bin spacing, the findings confirm that LTI frequency-response analysis reliably predicts practical analogue filter performance in anti-aliasing and noise-conditioning applications.

References

Edith Cowan University School of Engineering. (2026a). *ENS3553/ENS6160 Signals and Systems: Module 5 and Module 6 lecture slides* [Lecture slides]. Canvas.

Edith Cowan University School of Engineering. (2026b). *ENS3553/ENS6160 Signals and Systems: Tutorial 5 Question 6 and Laboratory Experiment 2 instruction sheet* [Course handout]. Canvas.

Harris, F. J. (1978). On the use of windows for harmonic analysis with the discrete Fourier transform. *Proceedings of the IEEE*, *66*(1), 51–83. https://doi.org/10.1109/PROC.1978.10837

Horowitz, P., & Hill, W. (2015). *The art of electronics* (3rd ed.). Cambridge University Press.

Keysight Technologies. (2014). *33220A 20 MHz function/arbitrary waveform generator: Data sheet* (Publication No. 5988-8544EN). Keysight Technologies.

Keysight Technologies. (2022). *InfiniiVision 4000 X-Series oscilloscopes: Data sheet* (Publication No. 5991-1103EN). Keysight Technologies.

Lathi, B. P., & Green, R. A. (2018). *Linear systems and signals* (3rd ed.). Oxford University Press.

Oppenheim, A. V., Willsky, A. S., & Nawab, S. H. (1997). *Signals and systems* (2nd ed.). Prentice Hall.

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